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In [1]: import pandas as pd
import numpy as np
from sklearn.preprocessing import LabelEncoder
from sklearn.preprocessing import StandardScaler, MinMaxScaler
from sklearn.model_selection import train_test_split
df = pd.read_csv("cleaned_pbl_dataset.csv")
print("===== DATASET LOADED SUCCESSFULLY =====")
print("Rows:", df.shape[0], "Columns:", df.shape[1])

#IDENTIFY CATEGORICAL COLUMNS
categorical_columns = df.select_dtypes(include='object').columns
print("\n===== CATEGORICAL COLUMNS =====")
print(categorical_columns)

#LABEL ENCODING
label_encoder = LabelEncoder()
for column in categorical_columns:
    df[column] = label_encoder.fit_transform(df[column])
print("\n===== LABEL ENCODING COMPLETED =====")
print(df.head())

#SEPARATE FEATURES AND TARGET
X = df.drop("Label", axis=1)
y = df["Label"]
print("\n===== FEATURES AND TARGET CREATED =====")
print("Feature Columns:", X.columns)

#STANDARD SCALING
standard_scaler = StandardScaler()
X_standard_scaled = standard_scaler.fit_transform(X)
print("\n===== STANDARD SCALING COMPLETED =====")

#MIN MAX SCALING
minmax_scaler = MinMaxScaler()
X_minmax_scaled = minmax_scaler.fit_transform(X)
print("===== MIN MAX SCALING COMPLETED =====")

#TRAIN TEST SPLIT
X_train, X_test, y_train, y_test = train_test_split(
    X_standard_scaled,
    y,
    test_size=0.20,
    random_state=42
)
print("\n===== TRAIN TEST SPLIT COMPLETED =====")
print("X_train Shape:", X_train.shape)
print("X_test Shape:", X_test.shape)
print("y_train Shape:", y_train.shape)
print("y_test Shape:", y_test.shape)

scaled_df = pd.DataFrame(X_standard_scaled, columns=X.columns)

scaled_df["Label"] = y.values

scaled_df.to_csv("preprocessed_pbl_dataset.csv", index=False)

print("\n===== PREPROCESSING COMPLETED SUCCESSFULLY =====")
print("Preprocessed dataset saved as: preprocessed_pbl_dataset.csv")

```

===== DATASET LOADED SUCCESSFULLY =====

Rows: 1594 Columns: 12

===== CATEGORICAL COLUMNS =====

Index(['Protocol', 'Encryption_Type'], dtype='object')

===== LABEL ENCODING COMPLETED =====

	Packet_Size	Connection_Duration	Error_Rate	Protocol	Source_Port \
0	402	5	0.20	1	10459
1	735	1	0.84	0	38790
2	570	5	0.82	2	1186
3	406	5	0.64	1	35766
4	371	2	0.25	1	20699

	Destination_Port	Bytes_Sent	Bytes_Received	Login_Attempts \
0	20	18090	4422	1
1	53	7350	10290	9
2	20	8550	3990	8
3	25	17052	2030	1
4	143	3710	4823	1

	Failed_Logins	Encryption_Type	Label
0	1	3	0
1	3	0	1
2	3	3	1
3	1	3	0
4	1	3	0

===== FEATURES AND TARGET CREATED =====

Feature Columns: Index(['Packet_Size', 'Connection_Duration', 'Error_Rate', 'Protocol', 'Source_Port', 'Destination_Port', 'Bytes_Sent', 'Bytes_Received', 'Login_Attempts', 'Failed_Logins', 'Encryption_Type'], dtype='object')

===== STANDARD SCALING COMPLETED =====

===== MIN MAX SCALING COMPLETED =====

===== TRAIN TEST SPLIT COMPLETED =====

X_train Shape: (1275, 11)

X_test Shape: (319, 11)

y_train Shape: (1275,)

y_test Shape: (319,)

===== PREPROCESSING COMPLETED SUCCESSFULLY =====

Preprocessed dataset saved as: preprocessed_pbl_dataset.csv

In []: